

RESPONSIBLE ML AUDIT

HMDA 2024 Mortgage Approval Model

Predicting loan approval from the 2024 HMDA Loan Application Register

Group 6 | DNSC 6330 Responsible Machine Learning | GWU | May 2026

Presented to the Internal Review Board

Measure

Diagnose

Document

Audit Framework: Five Questions

Q1

Optimization Objective

Q2

Known Failure Modes

Q3

Subgroup Error Measurement

Q4

Residual Risks After Mitigation

Q5

Is Deployment Defensible?

HMDA 2024 — Dataset at a Glance

17M+

Total Applications

8.66M

After Filtering

74.4%

Approval Rate

50 States

Geographic Scope

Label Construction

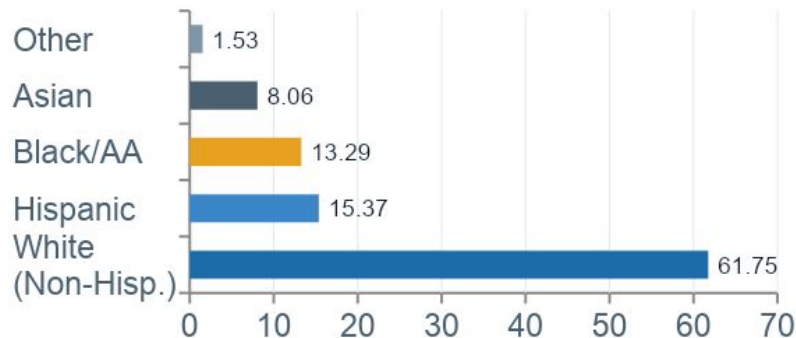
action_taken = 1 → Label 1 (Approved)

action_taken = 2 → Label 1 (Approved)

action_taken = 3 → Label 0 (Denied)

4, 5, 6, 7, 8 → Excluded

Race Breakdown



What Is the Model Optimizing For?

Primary Metric: AUC-ROC

- AUC measures discrimination across all thresholds
- More robust than accuracy under 74/26 class imbalance
- Threshold = 0.80 to reduce false approvals while monitoring subgroup FNRs
- Business rationale: lenders need a model that separates likely approvals from denials better than accuracy alone



Known Tradeoff

Optimizing overall AUC can hide poor subgroup performance. GBT scores 0.838 overall but only 0.815 for Black applicants — a gap that carries real-world consequences.

Who Bears the Cost?

Applicants

Denied access to credit if FPR is high for minority groups

Lenders

Regulatory risk if model violates ECOA or Fair Housing Act

Regulators

Systemic fair-lending violations if AIR < 0.80 not flagged

Community

Redlining if geographic proxies encode race implicitly

**Protected Attributes**

Race, Sex, Applicant Age, Sex Code

Featured used in Modeling

Feature	Definition
Income	The gross annual income, in thousands of dollars, relied on in making the credit decision, or if a credit decision was not made, the gross annual income relied on in processing the application
Property Value	The value of the property securing the covered loan or, in the case of an application, proposed to secure the covered loan, relied on in making the credit decision
Debt-to-income Ratio	The ratio, as a percentage, of the applicant's or borrower's total monthly debt to the total monthly income relied on in making the credit decision
ffies_msa_md_median_family_income	FFIEC Median family income in dollars for the MSA/MD in which the tract is located
Loan Purpose	The purpose of covered loan or application
Reverse Mortgage	Whether the covered loan or application is for a reverse mortgage
Submission_of_application	Whether the applicant or borrower submitted the application directly to the financial institution
Business_or_commercial_purpose	Whether the covered loan or application is primarily for a business or commercial purpose
Occupany_type	Occupancy type for the dwelling
Construction_method	Construction method for the dwelling

Where the Model Is Expected to Fail

Failure Type	Trigger Condition	Affected Group	Severity
Subgroup underperformance	Fewer minority training examples; higher variance	American Indian, Native Hawaiian	HIGH
Input gaming (evasion)	Applicant manipulates income or DTI to cross threshold	Sophisticated applicants	MEDIUM
Label-flip data poisoning	Insider flips approval labels in training data	Black applicants targeted in lab sweep	MEDIUM
Age disparity	Applicants >74 are below the threshold of 0.8 relative to applicants aged 45-54	Elderly applicants (>74)	LOW-MED

SHAP & LIME: What Drives Predictions?

LIME — GBT Top Features

Black Applicant (P = 0.803)

loan_purpose_1 ≤ 0 **-0.174**

-0.66 < debt_to_income_ratio ≤ 0.06 **+0.117**

-0.01 < income ≤ 0.01 **+0.059**

-0.04 < property_value ≤ 0.03 **+0.043**

White Applicant (P = 0.813)

loan_purpose_1 ≤ 0 **-0.161**

0.06 < debt_to_income_ratio ≤ 0.61 **+0.108**

income > 0 **+0.042**

property_value > 0.03 **-0.02**

Key Governance Findings

Same top features, different weights

Black and White applicants share the same top driver (loan purpose) but DTI and property value weights diverge, suggesting different decision logic by group.

DTI is the dominant risk signal

The most denied applicants in both groups have DTI > 0.61, which LIME confirms contributes -0.37 to approval probability — the single largest feature impact.

Proxy risk: tract_minority_population

This geographic feature correlates strongly with race and appears in SHAP global importance. Including it as a model feature constitutes potential disparate treatment.

DiCE: What Would Flip a Denial?

DiCE generates counterfactuals: the minimum input changes needed to flip a model's decision. We ran DiCE on the GBT model varying income, DTI, property value, and MSA median family income.

Black Applicant — Approved (P=0.938)

N/A — Already
Approved

Model approved at high confidence. No counterfactual needed — DTI in favorable range (0.06–0.61), purchase loan.

Black Applicant — Denied (P=0.037)

Reduce DTI or
increase income

Denial driven by DTI > 0.61 (−0.37 LIME weight) and low income. Counterfactual: reduce DTI to 40.4% or increase income by ~55k.

White Applicant — Approved (P=0.940)

N/A — Already
Approved

High confidence approval. Same feature profile structure as Black approved applicant — DTI and loan purpose are the key drivers.

White Applicant — Denied (P=0.035)

Reduce DTI or
increase income

Same denial drivers as Black denied applicant. DTI > 0.61 and income below threshold. Counterfactuals require similar magnitude of change.

Adversarial Attack Surface

MEDIUM

Evasion (Input Gaming)

Evasion (Input Gaming)

Adversary: Applicant | Method: Manipulate income or DTI to cross 0.80 threshold

Finding: GBT: epsilon=0.1 shifts mean approval prob by -0.0054. LR more sensitive to gradient-based perturbation.

HIGH

Label-Flip Poisoning

Label-Flip Poisoning

Adversary: Insider / Data Handler | Method: Flip approved labels for Black applicants in training data

Finding: GBT: 5% poison rate produces delta AUC of only +0.0004 while fairness metrics shift materially — invisible to AUC monitoring

LOW

Membership Inference

Membership Inference

Adversary: API User | Method: Query model to determine if specific applicant was in training data

Finding: GBT MI AUC = 0.500, LR MI AUC = 0.500, no detectable privacy leakage. Near-zero gen gaps prevent exploitation.

Three Models Compared — GBT Selected

Baseline	SELECTED	Challenger
Logistic Regression	Gradient Boosting	Random Forest
Test AUC	Test AUC	Test AUC
0.781	0.838	0.843
Brier Score	Brier Score	Brier Score
0.145	0.122	0.121
AIR (approval)	AIR (approval)	AIR (approval)
0.923	0.945	0.938
Gen Gap	Gen Gap	Gen Gap
0.000	0.001	0.001
FPR Black	FPR Black	FPR Black
0.254	0.206	0.236
FPR White	FPR White	FPR White
0.239	0.184	0.254
Interpretable but lowest AUC; highest Black FPR.	GBT selected: highest AUC (0.838), lowest Brier (0.122), and lowest Black FPR (0.206), but residual subgroup gaps remain.	Strong Brier and gen gap, but higher Black FPR than GBT

GBT selected: strong AUC and Brier, lowest Black FPR, but subgroup gaps remain.

Disparate Impact Audit: Race (GBT Model)

Group	N (Test)	Approval Rate	AIR vs White	FPR	FNR	AUC
White (Non-Hispanic)	488,758	53.8%	Reference	0.184	0.359	0.833
Black / African American	105,564	50.8%	0.945	0.206	0.320	0.833
Hispanic or Latino	121,967	58.8%	1.093	0.220	0.257	0.847
Asian	63,939	70.3%	1.308	0.258	0.174	0.864
Am. Indian / AK Native	9,432	47.9%	0.891	0.158	0.340	0.849

AIR = group protected group approval rate ÷ reference group approval rate .

*Note: Four-fifths rule:
AIR < 0.80 is flagged*

Black AIR = 0.945, above the 0.80 threshold. However, Black FPR is higher than White FPR by 2.2pp and statistically significant.

Age >74 AIR = 0.898, above 0.80 but still lower than the 45–54 reference group. The main concern is the >74 female subgroup, where AIR falls below 0.80 . GBT age intersectional worst group: >74 Female at AIR 0.623. Requires ECOA review.

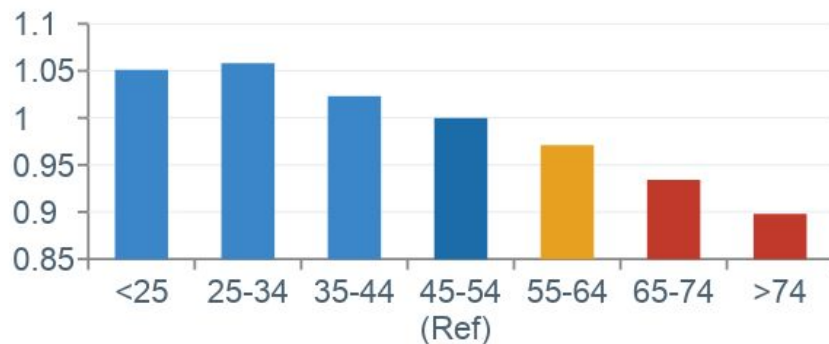
Disparate Impact Audit: Sex and Age

By Sex (GBT Model)



Female AIR = 0.958 (above 0.80 threshold). SMD = -10.11.
Statistically significant but not practically flagged under current guidelines.

By Age Group (GBT Model — ref: 45-54)



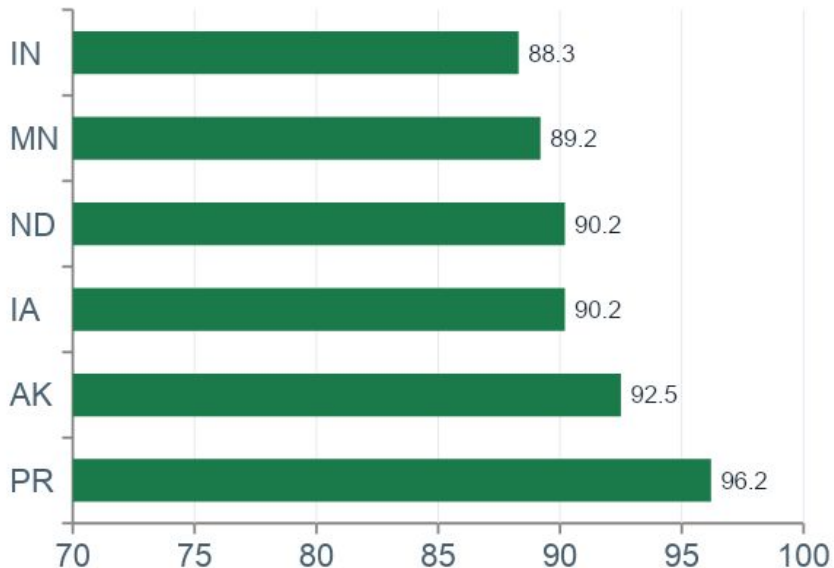
Applicants aged >74 have AIR = 0.898 and SMD = -23.8 (practically significant). ECOA protects against age discrimination — this warrants investigation.

Intersectional Analysis (Race × Sex)

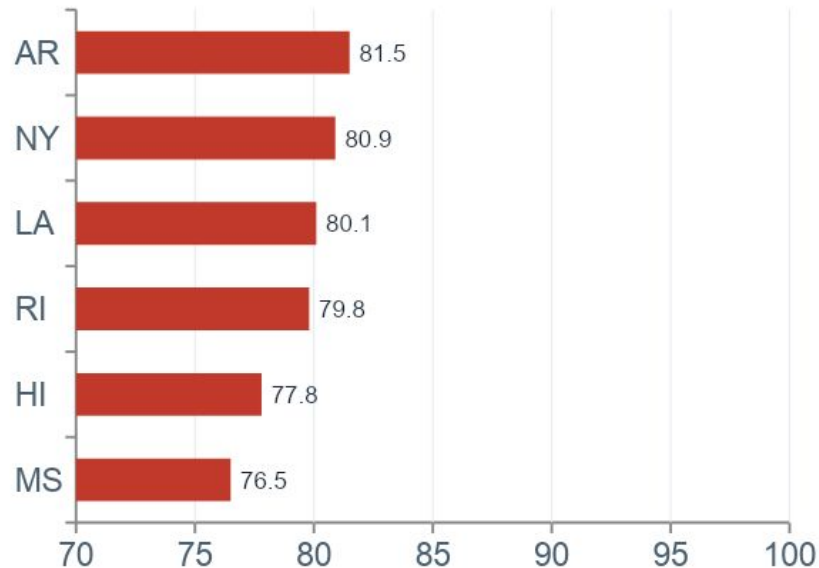
Intersectional audit confirmed disparities invisible in single-attribute analysis. Combined race and sex subgroups reveal concentrated harm for specific applicant profiles that aggregate metrics would mask.

State-Level Approval Rate Disparities

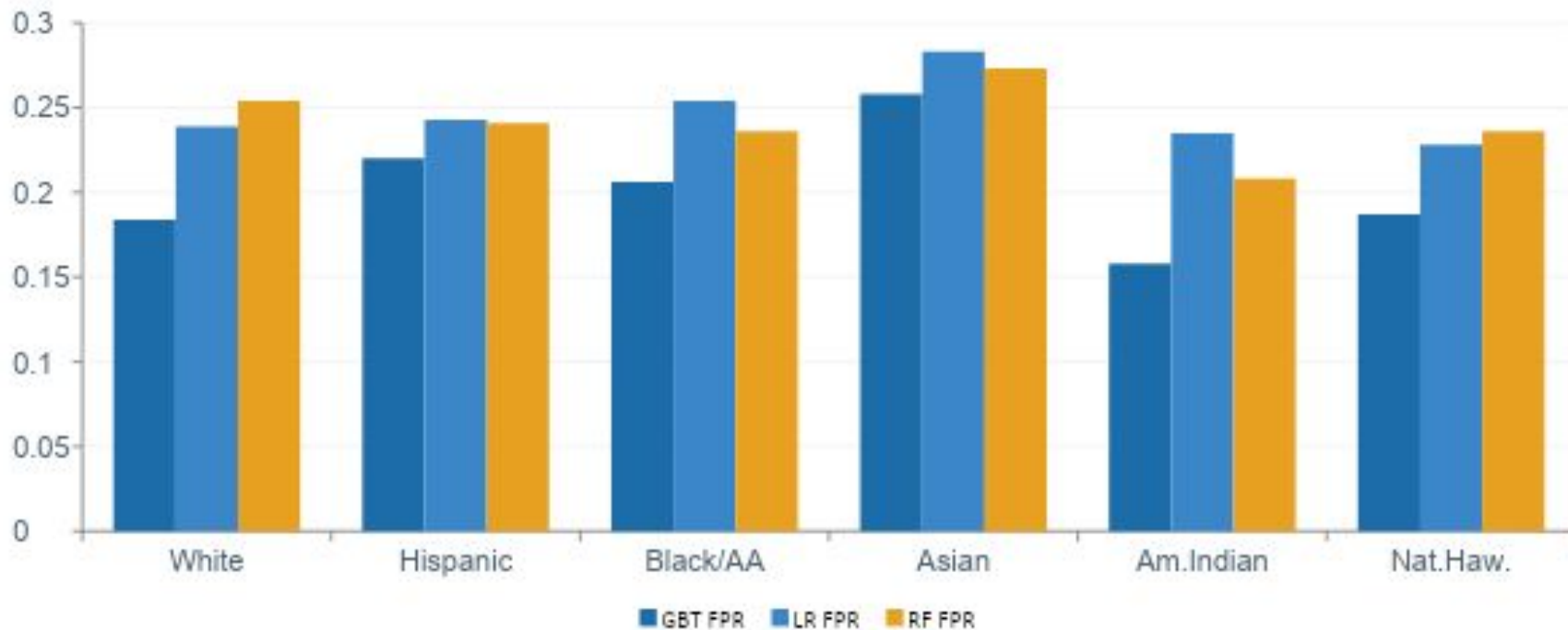
Highest Approval Rate States



Lowest Approval Rate States



All Three Models: FPR by Race



GBT has the lowest FPR for Black/AA applicants (0.206) across all three models. RF has the lowest for Am. Indian (0.208). LR is highest across most groups.

Model Stability Under Stress and Drift

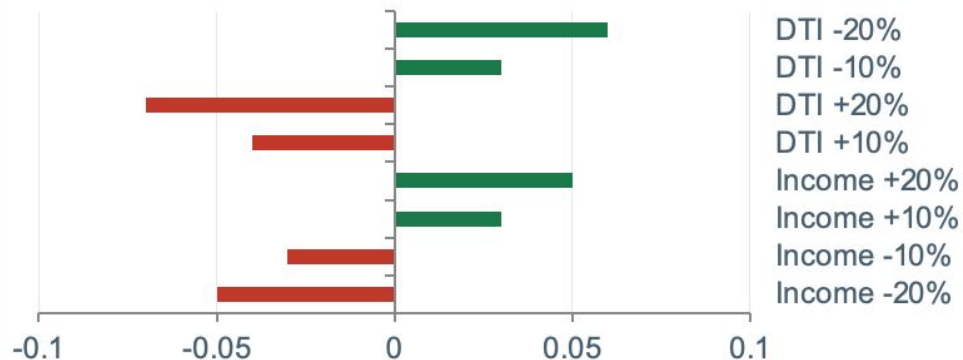
0.0006GBT Gen Gap
(Train-Test AUC)**-0.0003**

LR Gen Gap

0.0008RF Gen Gap
(pending)**Stable**

PSI — Key Features

Stress Test: Income & DTI Sensitivity (GBT)



Drift Assessment

- GBT and LR gen gaps near zero — no overfitting detected
- All input features show PSI = 0.0 (Stable) — no covariate shift
- DTI most sensitive: avg prediction range 0.73 vs income 0.03
- Interest rate environment is primary post-2024 concept drift risk
- GBT Brier = 0.122, LR Brier = 0.145 — GBT better calibrated

What Remains After Mitigation?

Risk	Mitigation Attempted	Outcome	Accepted Residual
Racial FPR disparity (GBT Black FPR = 0.206 vs White 0.184)	Model selection (GBT outperforms LR/RF on Black FPR); threshold tuning	GBT lowest Black FPR of three models. Gap reduced but not eliminated	Small residual gap accepted. Monitor post-deployment monthly
Proxy discrimination (tract_minority_population_percent)	Feature documented; governance flag added to audit record	Feature retained — predictive value vs harm under review	Requires business necessity justification before deployment
Age disparity >74 (AIR below 0.80 threshold)	Flagged in audit. No model change — ECOA compliance under review	Practically significant. Intersectional worst group AIR = 0.490	Cannot accept — requires legal review before deployment
Concept drift post-2024	PSI and KS monitoring plan; retraining triggers defined	No current drift detected. All PSI = 0.0 (Stable). Forward-looking risk	Accepted with quarterly recalibration schedule
Label-flip poisoning vulnerability	Subgroup FPR monitoring added; data provenance controls recommended	Attack confirmed invisible to AUC (delta = 0.0004 at 5% poison rate)	Residual insider threat — data access controls required

Post-Deployment Monitoring Framework

Area	Metric	Alert Trigger	Frequency
Fairness	AIR by race and sex	AIR < 0.80 or > 1.25	Monthly
Fairness	FPR disparity by race	FPR gap > 5pp	Monthly
Performance	Test AUC and log loss	AUC drops > 2pp from baseline	Monthly
Calibration	Brier score by subgroup	Subgroup Brier > 0.20	Quarterly
Drift	PSI on income, DTI, property value	PSI > 0.25 on any feature	Monthly
Security	Subgroup FPR post-retrain	Any > 10% shift from prior	Each retrain

Escalation policy: any fairness alert triggers a 30-day remediation review. Unresolved AIR violations pause deployment. GitHub repo maintained as the audit record.

Is Deployment Defensible?

Recommendation: CONDITIONAL DEPLOYMENT — GBT Model with Active Monitoring

Defensible — Yes

- Optimization objective documented with business rationale
- Failure mode table with trigger conditions and severity
- AIR, SMD, FPR disparity computed across race, sex, age, state
- Intersectional analysis completed
- Robustness and drift testing conducted
- Monitoring plan with defined alert thresholds
- Reproducible GitHub codebase as audit record

Conditions Required Before Full Deployment

- Legal review of >74 female AIR below 0.80 under ECOA
- Business necessity justification for tract_minority_population_percent
- Data access controls to prevent insider label-flip poisoning
- RF challenger results documented, but GBT retained due to lower Black FPR
- Threshold 0.80 validated against lender's risk appetite

Thank You

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GitHub Repository: <https://github.com/bek-92/hmda-responsible-ml-audit>

Q1 Optimization Objective

Q2 Known Failure Modes

**Q3 Subgroup Error
Measurement**

Q4 Residual Risks

**Q5 Is Deployment
Defensible?**

A good audit is not a perfect model. It is a justified model with documented risks.

Disparate Impact Audit: Sex (GBT Model)

Group	Reference Group	Group Category	Observations	Percent Missing	Total	Favorable	Percent Favorable	Percent Difference Favorable	AIR	P-Values	Practically Significant	Shortfall
is_Female	is_Male	sex	792493	0.0	319170	170150	0.533101	0.034952	0.93847	6.015546e-207	No	NaN
is_Male		sex	792493	0.0	473323	268873	0.568054	NaN	NaN	NaN		NaN

Disparate Impact Audit: Age (GBT Model)

Group	Reference Group	Group Category	Observations	Percent Missing	Total	Favorable	Percent Favorable	Percent Difference Favorable	AIR	P-Values	Practically Significant	Shortfall
is_25-34	is_45-54	age	792493	0.0	160916	115592	0.718338	-0.209808	1.412579	0.000000e+00	No	NaN
is_35-44	is_45-54	age	792493	0.0	191809	114297	0.595890	-0.087361	1.171791	0.000000e+00	No	NaN
is_55-64	is_45-54	age	792493	0.0	132772	59394	0.447338	0.061191	0.879671	1.626504e-245	No	NaN
is_65-74	is_45-54	age	792493	0.0	73170	28070	0.383627	0.124902	0.754386	0.000000e+00	Yes	9139.073106
is_<25	is_45-54	age	792493	0.0	33130	24977	0.753909	-0.245380	1.482528	0.000000e+00	No	NaN
is_>74	is_45-54	age	792493	0.0	29986	9882	0.329554	0.178975	0.648053	0.000000e+00	Yes	5366.753125
is_45-54		age	792493	0.0	170710	86811	0.508529	NaN	NaN	NaN		NaN

Disparate Impact Audit: Race and Sex (GBT Model)

Group	Reference Group	Group Category	Observations	Percent Missing	Total	Favorable	Percent Favorable	Percent Difference Favorable	AIR	P-Values	Practically Significant	Short fall
is_Native Hawaiian or Other Pacific Islander / Female	is_White (Non-Hispanic) / Male	subgroup	792493	0.0	1034	458	0.442940	0.108365	0.803439	3.352622e-12	No	NaN
is_Native Hawaiian or Other Pacific Islander / Male	is_White (Non-Hispanic) / Male	subgroup	792493	0.0	1799	831	0.461923	0.089382	0.837873	3.596517e-14	No	NaN
is_American Indian or Alaska Native / Female	is_White (Non-Hispanic) / Male	subgroup	792493	0.0	3598	1682	0.467482	0.083823	0.847955	1.128493e-23	No	NaN
is_American Indian or Alaska Native / Male	is_White (Non-Hispanic) / Male	subgroup	792493	0.0	5834	2839	0.486630	0.064675	0.882688	9.207491e-23	No	NaN
is_Black or African American / Female	is_White (Non-Hispanic) / Male	subgroup	792493	0.0	56156	28205	0.502262	0.049043	0.911041	1.572744e-101	No	NaN

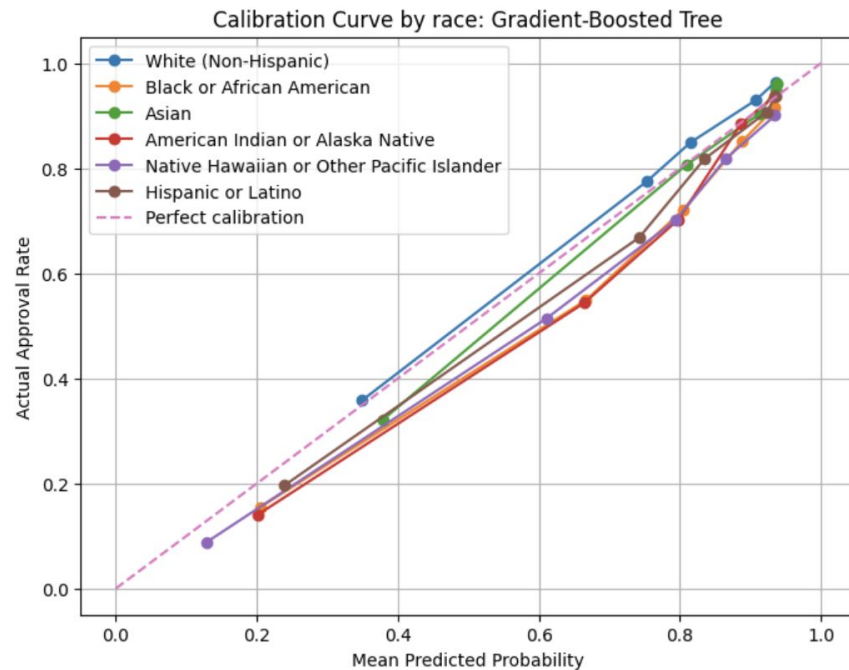
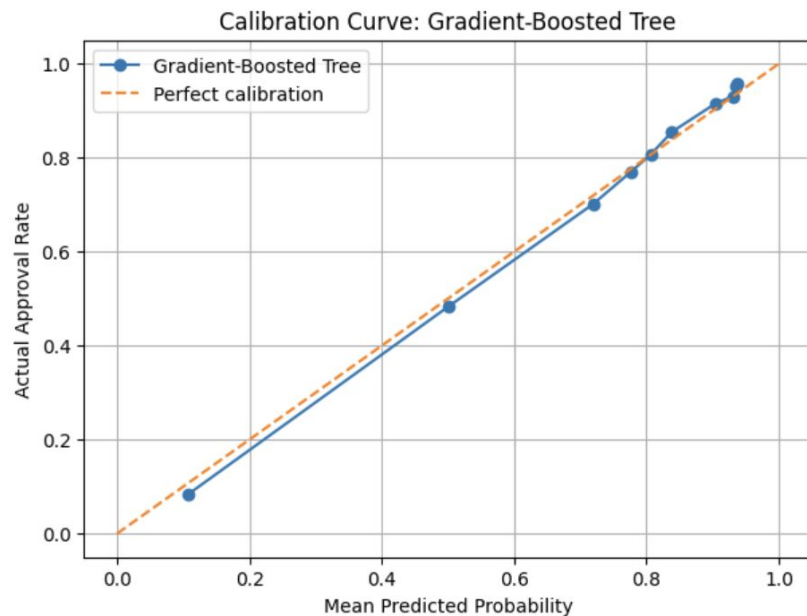
Disparate Impact Audit: Age and Sex (GBT Model)

Group	Reference Group	Group Category	Observations	Percent Missing	Total	Favorable	Percent Favorable	Percent Difference Favorable	AIR	P-Values	Practically Significant	Shortfall
is_>74_Female	is_45-54_Male	subgroup	792493	0.0	16178	5180	0.320188	0.193709	0.623058	0.000000e+00	Yes	3133.828175
is_>74_Male	is_45-54_Male	subgroup	792493	0.0	13808	4702	0.340527	0.173370	0.662637	0.000000e+00	Yes	2393.891918
is_65-74_Female	is_45-54_Male	subgroup	792493	0.0	36291	13922	0.383621	0.130276	0.746494	0.000000e+00	Yes	4727.841656
is_65-74_Male	is_45-54_Male	subgroup	792493	0.0	36879	14148	0.383633	0.130264	0.746517	0.000000e+00	Yes	4804.013183

Disparate Impact Audit: Generalization Gap (GBT Model)

	dataset	accuracy	auc	log_loss
0	Train	0.7087	0.8383	0.3942
1	Test	0.7080	0.8377	0.3950
2	Train - Test Gap	0.0006	0.0006	0.0007

Calibration (GBT Model)



ICE Curves (GBT Model)

